

Flowcharts

visual algorithms

Action

Represented in *squares*



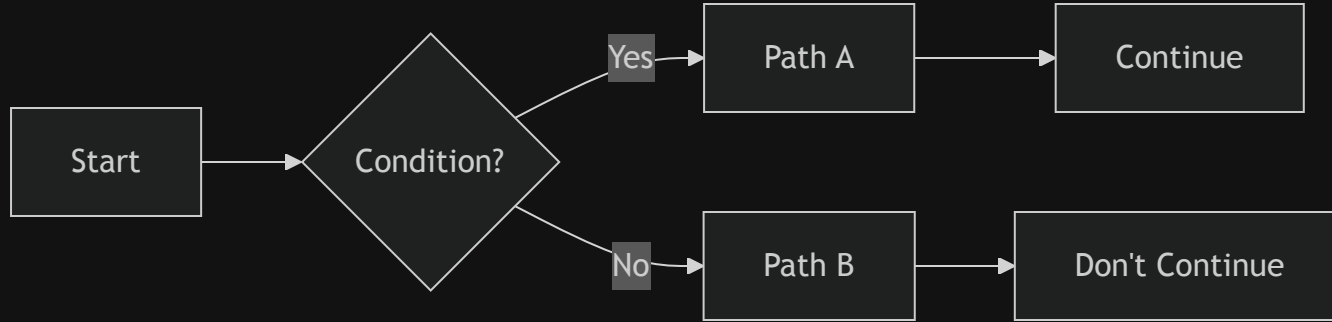
Flowcharts take advantage of the fact that humans are *much better* at processing *visual* information, than textual

Remember the Robot Maze

Given the `robot` and maze from the previous lesson, make a flowchart to get it from point A to point B

Decisions

Represented using *diamonds*



Almost *every single program* is simply a combination of *steps* and *decisions*

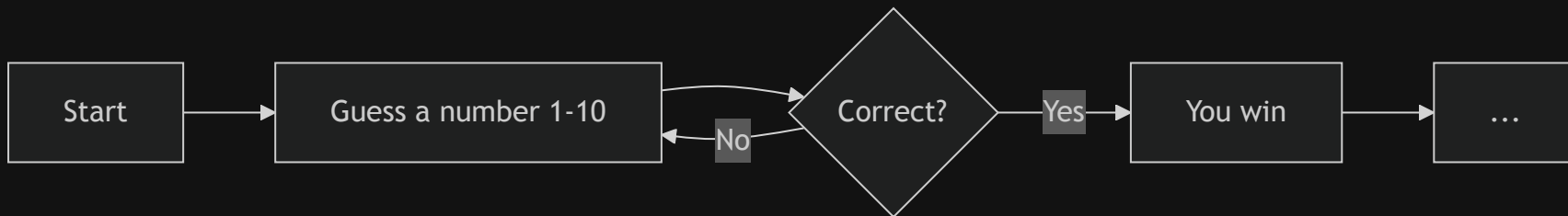
If you can find a problem, and break it down into a series of *steps and decisions*, then you can make a program for it

Recommendation flowcharts

Loops

Represented using a *combination* of decisions and steps, characterized by

- a visible *loop*
- a decision point for starting or stopping the loop



Back to the maze (again)

Assume the `robot` has an extra function called

- `check if front is obstructed` and the robot will say *yes* if it is, and *no* if it isn't

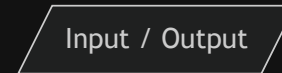
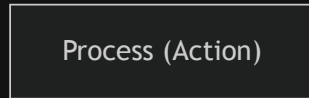
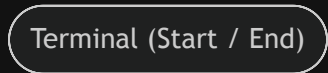
Create a flowchart that solves the maze with **only one** `move forward` action block being used

So the possible actions are

- check if front is obstructed
- check if left is obstructed
- check if right is obstructed
- check if back is obstructed
- turn right 90 degrees
- turn left 90 degrees
- move forward (limited to 1)

Other shapes

Shape	Name	Symbol
Oval	Start / End	()
Rectangle	Action / Process	[]
Diamond	Decision	{ }
Parallelogram	Input / Output	/ /
Cylinder	Data / Storage	[()]



Morning routine flowchart

build a flowchart for your morning routine

You can use [draw.io](#), [excalidraw](#), [paint](#), [paper](#), etc

Sorting algorithm flowchart

build a flowchart that sorts 5 jumbled numbers

You're possible moves are

- select the first number
- move the selection left
- check current number
- check the number after the current
- compare values
- swap
- check if sorted

From

1 51428 → 14258